

Quantum Biology Atlas: evidence-tiered pathway maps for magnetic-field effects in plants

Richard Barker*

University of Wisconsin–Madison

Abstract

Summary: Claims about magnetic-field effects in biology are easy to draw and hard to qualify: a pathway diagram gives a hypothesised radical-pair site and a measured one the same visual weight. The Quantum Biology Atlas separates the two questions that are routinely conflated — what quantum chemistry a molecular entity carries, and how much is actually known about its field sensitivity — and renders the second as a border style, so a hypothesis looks provisional on the page. It publishes 10 identifier-bound pathway maps (126 annotated nodes, 168 edges, 125 *Arabidopsis* loci) as SBGN-ML [1] with the quantum annotation in extension blocks, a Python package that projects measured expression onto them across 5 species, and a browser tool that does the same for a user’s own table without uploading it. The overlay machinery refuses rather than degrades: a join that matches nothing raises instead of rendering a pale map that looks like a result. Applied to its own corpus the atlas returns two negatives that constrain how its figures may be read, and reports both.

Availability and implementation: Source, ontology, maps and evidence base at <https://github.com/dr-richard-barker/quantum-biology-atlas>; interactive maps and the upload tool at <https://dr-richard-barker.github.io/quantum-biology-atlas/>. Code is MIT; ontology, maps and evidence base are CC-BY-4.0.

Contact: dr.richard.barker@gmail.com

1 Introduction

The proposal that weak magnetic fields act on biology through spin-correlated radical pairs is specific, testable and now well developed theoretically [2, 3]. In plants it has a named candidate receptor, cryptochrome [4–7], and a substantial phenomenology: near-null fields alter flowering time, hypocotyl growth, reproductive yield and mineral nutrition [8–12].

The gap is between those two bodies of work. The mechanism literature is about chemistry; the phenomenology is about phenotypes; and pathway diagrams drawn to connect them do not distinguish a site where a field effect has been *measured* from one where a plausible radical intermediate merely exists. Both appear as a box with an arrow. That is not a presentation problem — it is how a hypothesis acquires the appearance of an established route.

The atlas addresses this narrowly. It does not adjudicate the mechanism. It makes the evidence state of every node explicit, machine-readable and visible, and it makes projecting real data onto those nodes the only way a number ever appears on a figure.

2 Implementation

2.1 Two axes, kept orthogonal

The ontology (QBO, version 0.1.0, 83 entities) annotates each entity on two independent axes. `quantum_class` records the chemistry it carries — a flavin semiquinone, a [2Fe-2S] or [4Fe-4S] cluster, a triplet ground state,

*Correspondence: dr.richard.barker@gmail.com

a site of reactive-oxygen leak. `evidence_tier` records how much is known about its magnetic-field sensitivity: T1 demonstrated, T2 inferred, T3 plausible, T4 context.

Collapsing these is the failure the design exists to prevent, and the validator enforces their independence in both directions. A tier that asserts a demonstrated or inferred field effect cannot validate without a DOI resolved against CrossRef — the evidence base holds 42 such references, each title-matched against the source —; a T3 node cannot validate without a written rationale. Conversely, an entity classed as structural context *may* carry a demonstrated effect — iron uptake under near-null fields is a measured phenomenon with no proposed mechanism [12, 13] — and the validator requires only that the tension be stated, not that it be resolved.

2.2 Tier as a visual channel

Evidence tier is rendered as the node’s border: heavy solid for T1, light solid for T2, dashed for T3, hairline for T4. The consequence is that the atlas’s own maps look mostly provisional, because they are: of 126 nodes, 4 are T1, 31 are T2, 44 are T3 and 47 are T4 — 72 % carry no measured field effect at all. That is the finding rather than a defect. The near-null-field literature is developmental and nutritional, and has not yet measured a respiratory complex directly.

Maps are declarative sources carrying no coordinates; geometry is derived from measured text metrics, so a label cannot outgrow its box. Rendering is checked by a probe that runs in a real browser and asserts on what was painted — no overlapping text, nothing clipped, every overlay mark on its own node — using the browser’s own text layout rather than re-running the compiler’s arithmetic.

2.3 Identifiers, which are where the joins actually fail

Four of the six demonstrations needed a different identifier route, and each wrong choice fails silently rather than loudly. Published supplementary tables write *Arabidopsis* loci as `At1g01980.1`, which a conventional AGI pattern matches zero times. A two-colour array platform’s `GENE_SYMBOL` column holds a locus only sometimes — often a trivial name instead — and reading it alone reaches 17 of the atlas’s 125 loci where the accession field reaches all of them. Main-text tables are keyed on gene symbols, which cannot safely be resolved from memory: authoring this ontology turned up five symbol collisions that would each have put a different gene on a map, so symbols are resolved from each paper’s own primer table.

2.4 Projection, and what it refuses

Measured values reach a figure only through `qbio.project`, from a named study and a named contrast, with the provenance travelling into the caption. Three refusals are enforced rather than warned:

- A join matching no node **raises**. A pale map rendered from nothing looks like a result, which is worse than an error.
- Coverage below 25 % raises unless explicitly overridden, and the coverage is stated in the caption either way.
- A series still on a ratio or signed-fold scale is refused. Published tables use at least three conventions, and feeding a ratio to a $\log_2$ renderer colours every unchanged gene as strongly upregulated while looking entirely normal.

Cross-species projection runs two independent backbones — Ensembl Compara `pan_homology` [14] and a frozen OrthoDB matrix [15] — and reports where they disagree rather than preferring one. 68 of 125 loci reach human, of which 22 are called by both methods; 63 reach *Drosophila*, where only Ensembl can reach — the committed matrix is human-anchored, and the coverage report says so rather than letting one method read as two. Unmapped loci are listed, never dropped: the alternative oxidases fail to map to human because humans do not have one, and that is a result.

2.5 Time, and the limit of a node

A time course is carried intact to the renderer and collapsed to one colour only at the last moment, by largest-magnitude rather than mean: a gene that rises early and falls late averages to “no response”. Single-locus nodes carry a sparkline of the trajectory. Multi-locus nodes do not — they carry a per-locus heatmap instead, one row per locus and one column per timepoint (Fig. 1D).

That choice was forced by a false positive. Under ionising radiation the alternative oxidase node appeared to respond in the same direction as under a near-null field, a clean shared result. At locus level AOX1D rises under both while AOX2 *falls* under radiation, and the aggregate was selecting a different family member at each timepoint. Almost every multi-locus node behaves this way under a strong perturbation, so `project_series` now records per-timepoint sign divergence and the caption says which nodes have it. A gene family is not a unit of response.

2.6 Browser tool

`explore.html` performs the same join client-side, taking the shape of ShinyGO [16] but for hand-curated quantum-annotated maps rather than enrichment. The file is read in the browser and never uploaded — unpublished data should not have to leave the machine to be looked at — and the refusals above are ported intact and rendered on the page, because a refusal that reaches only the console is not a refusal.

3 Example

3.1 Demonstrations

Six demonstrations span three provenance classes, which the atlas labels separately because they do not carry equal weight: NASA GeneLab-processed differential expression [17], depositors’ normalised ratios with no model fitted, and values read from a paper’s tables where nothing was deposited. The last class is necessary because the group that produced most of the atlas’s T1/T2 evidence deposited no raw data, and each such page quotes the article’s Data Availability Statement verbatim.

Several are worth naming. OSD-27 places a plant-anchored ontology on *Drosophila* in a 16.5 T magnet: 63 loci reach the fly, and *Arabidopsis* CRY1 lands on FBgn0025680 — the canonical animal magnetoreceptor. Of 420 contrasts, 5 isolate field from the effective gravity that diamagnetic levitation changes alongside it, and across those, 0 of 15 nodes move consistently. Parmagnani *et al.*’s near-null time course [18] contributes 2,716 values across 194 loci at 7 timepoints in two tissues, and Agliassa *et al.*’s flowering series [10] adds 196 values across 26 loci, 7 of them atlas nodes — reaching the hormone/circadian map that no repository dataset does. Mannino *et al.*’s near-null study in sweet basil (*Ocimum basilicum*) [19] contributes 21 values across 21 loci, 4 of them atlas nodes on the redox map, demonstrating gene-metabolite decoupling in phenylpropanoid metabolism alongside divergent regulation of superoxide dismutase isoforms.

3.2 Two negatives the atlas reports about itself

The selected loci are not preferentially field-responsive. OSD-8 measures 29327 loci and contains the cleanest field-only contrast available: 16.5 T inside a magnet against 0 T outside it at matched effective gravity. The atlas’s loci move by mean $|\log_2\text{FC}|$ 0.1660 against 0.1763 for the same number of loci drawn at random ($p = 0.705$, 10,000 permutations). All seven experimental groups agree, including two with no magnet in them.

This does not invalidate an ontology that selects for mechanism rather than observed response, but it does constrain the figures: on this dataset, colour on a map is not evidence that a field acts selectively at these nodes. The pre-registered test on the corpus is likewise null (OR 1.58, $p = 0.48$) with a minimum detectable odds ratio of 4.0 at 80 % power — too underpowered to settle anything, which is reported rather than omitted.

The nodes are not inert, however. A 2-dose, 4-timepoint ionising-radiation series reaching 123 of the atlas’s loci moves the same loci up to 0.299 mean $|\log_2\text{FC}|$, roughly $1.8\times$ their response to the 16.5 T field, while remaining below their background ($p \geq 0.87$). So the magnetic null cannot be attributed to genes that do not move.

3.3 Overlap between perturbations

Radiation is not a magnetic-field regime and the atlas does not treat it as one; QBO’s field-regime vocabulary is not extended to it. But radiolysis makes radicals directly, so the question of overlap is worth asking. Of 11 loci measured under both ionising radiation and a near-null field — a locus counting as responding at $|\log_2\text{FC}| \geq 0.5$, applied identically to both sides so the overlap cannot be tuned — 4 respond to both and 2 agree in direction. That is reported as a table and explicitly not tested: below twenty shared units the permutation test refuses rather than returning a p -value that would read as evidence. Against the strong-field study, where 123 loci are shared and a test is possible, the observed overlap of 3 against 2.1 expected is unremarkable ($p = 0.352$).

4 Conclusion

The atlas is infrastructure for a field whose central claim is not settled, and its value is in making the unsettledness legible: which nodes carry which chemistry, what is actually known about each, and what a given dataset does and does not show. Its own two negatives are part of the output rather than an embarrassment — a tool that only reported confirmations would be the wrong tool for this literature.

Distribution is by PyPI, GitHub Pages and Zenodo rather than Bioconductor. The atlas is Python and TypeScript where Bioconductor is an R ecosystem; the rendering niche is already occupied by *SBGNview* and *pathview*; and ShinyGO, the tool this one is shaped after, is not a Bioconductor package either. A thin R annotation-data package exposing the QBO corpus to those renderers would be the novel layer without the competing one, and remains open.

Acknowledgements

The companion review [20] is in preparation and has no DOI; none is recorded here.

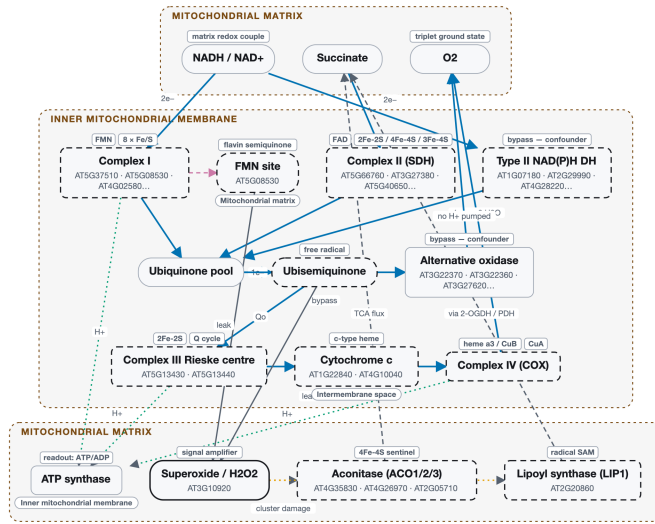
References

- [1] Nicolas Le Novère, Michael Hucka, Huaiyu Mi, Stuart Moodie, Falk Schreiber, Anatoly Sorokin, Emek Demir, Katja Wegner, Mirit I Aladjem, Sarala M Wimalaratne, Frank T Bergman, Ralph Gauges, Peter Ghazal, Hideya Kawaji, Lu Li, Yukiko Matsuoka, Alice Villéger, Sarah E Boyd, Laurence Calzone, Melanie Courtot, Ugur Dogrusoz, Tom C Freeman, Akira Funahashi, Samik Ghosh, Akiya Jouraku, Sohyoung Kim, Fedor Kolpakov, Augustin Luna, Sven Sahle, Esther Schmidt, Steven Watterson, Guanming Wu, Igor Goryanin, Douglas B Kell, Chris Sander, Herbert Sauro, Jacky L Snoep, Kurt Kohn, and Hiroaki Kitano. The Systems Biology Graphical Notation. *Nature Biotechnology*, 27(8):735–741, 2009. doi: 10.1038/nbt.1558.
- [2] Thorsten Ritz, Salih Adem, and Klaus Schulten. A Model for Photoreceptor-Based Magnetoreception in Birds. *Biophysical Journal*, 78(2):707–718, 2000. doi: 10.1016/S0006-3495(00)76629-X.
- [3] Hadi Zadeh-Haghighi and Christoph Simon. Magnetic field effects in biology from the perspective of the radical pair mechanism. *Journal of The Royal Society Interface*, 19(193), 2022. doi: 10.1098/rsif.2022.0325.
- [4] Margaret Ahmad, Paul Galland, Thorsten Ritz, Roswitha Wiltchko, and Wolfgang Wiltchko. Magnetic intensity affects cryptochrome-dependent responses in *Arabidopsis thaliana*. *Planta*, 225(3): 615–624, 2006. doi: 10.1007/s00425-006-0383-0.

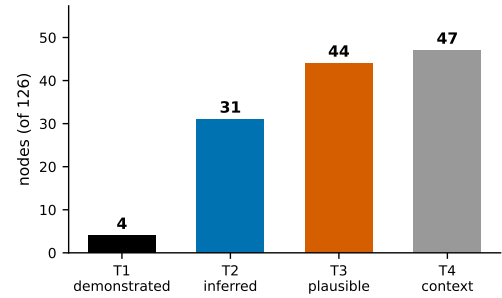
QBM-01 · Mitochondrial electron transport and oxidative phosphorylation

Magnetically addressable nodes of the plant respiratory chain, with the evidence tier for each shown as a border style

Evidence tier



(A)

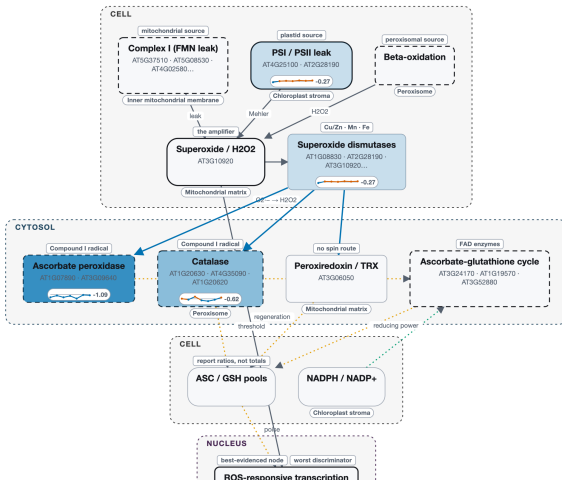


(B)

QBM-07 · ROS production, scavenging and redox buffering

The amplifier that makes the review's hypothesis testable — and the reason it cannot identify a mechanism on its own

Evidence tier

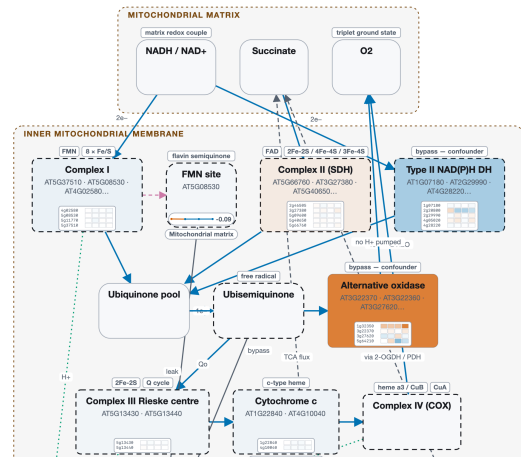


(C)

QBM-01 · Mitochondrial electron transport and oxidative phosphorylation

Magnetically addressable nodes of the plant respiratory chain, with the evidence tier for each shown as a border style

Evidence tier



(D)

Figure 1: The atlas and its outputs. (A) QBM-01, the mitochondrial electron transport map, with evidence tier as a border channel: heavy solid T1, light solid T2, dashed T3, hairline T4. Most of the map is dashed. (B) Distribution of the 126 nodes across tiers — 72 % are T3 or T4, which is the state of the literature rather than a limitation of the corpus. (C) A near-null-field time course [18] projected onto the redox map; single-locus nodes carry a sparkline of the trajectory. (D) The same machinery under ionising radiation, where multi-locus nodes carry a per-locus heatmap instead — one row per locus, one column per timepoint. This is what reveals a gene family splitting, which the aggregate hides. Every panel is a file the repository generated; none was drawn. The browser tool is described in Section 2.6 rather than shown, because illustrating it would mean a screenshot rather than an artefact.

- [5] Ilia A. Solov'yov, Danielle E. Chandler, and Klaus Schulten. Magnetic Field Effects in *Arabidopsis thaliana* Cryptochrome-1. *Biophysical Journal*, 92(8):2711–2726, 2007. doi: 10.1529/biophysj.106.097139.
- [6] Marootpong Pooam, Louis-David Arthaut, Derek Burdick, Justin Link, Carlos F. Martino, and Margaret Ahmad. Magnetic sensitivity mediated by the *Arabidopsis* blue-light receptor cryptochrome occurs during flavin reoxidation in the dark. *Planta*, 249(2):319–332, 2018. doi: 10.1007/s00425-018-3002-y.
- [7] Thawatchai Thoradit, Kanjana Thongyoo, Khwanchai Kamoltheptawin, Lalin Tunprasert, Mohamed A. El-Esawi, Blanche Aguida, Nathalie Jourdan, Kittisak Buddhachat, and Marootpong Pooam. Cryptochrome and quantum biology: unraveling the mysteries of plant magnetoreception. *Frontiers in Plant Science*, 14, 2023. doi: 10.3389/fpls.2023.1266357.
- [8] Chunxiao Xu, Xiao Yin, Yan Lv, Changzhe Wu, Yuxia Zhang, and Tao Song. A near-null magnetic field affects cryptochrome-related hypocotyl growth and flowering in *Arabidopsis*. *Advances in Space Research*, 49(5):834–840, 2012. doi: 10.1016/j.asr.2011.12.004.
- [9] Chunxiao Xu, Shufeng Wei, Yan Lu, Yuxia Zhang, Chuanfang Chen, and Tao Song. Removal of the local geomagnetic field affects reproductive growth in *Arabidopsis*. *Bioelectromagnetics*, 34(6):437–442, 2013. doi: 10.1002/bem.21788.
- [10] Chiara Agliassa, Ravishankar Narayana, Cinzia M. Berteau, Christopher T. Rodgers, and Massimo E. Maffei. Reduction of the geomagnetic field delays *Arabidopsis thaliana* flowering time through downregulation of flowering-related genes. *Bioelectromagnetics*, 39(5):361–374, 2018. doi: 10.1002/bem.22123.
- [11] Ravishankar Narayana, Judith Fliegmann, Ivan Paponov, and Massimo E. Maffei. Reduction of geomagnetic field (GMF) to near null magnetic field (NNMF) affects *Arabidopsis thaliana* root mineral nutrition. *Life Sciences in Space Research*, 19:43–50, 2018. doi: 10.1016/j.lssr.2018.08.005.
- [12] Monirul Islam, Massimo E. Maffei, and Gianpiero Vigani. The Geomagnetic Field Is a Contributing Factor for an Efficient Iron Uptake in *Arabidopsis thaliana*. *Frontiers in Plant Science*, 11, 2020. doi: 10.3389/fpls.2020.00325.
- [13] Gianpiero Vigani, Monirul Islam, Viviana Cavallaro, Fabio F. Nocito, and Massimo E. Maffei. Geomagnetic Field (GMF)-Dependent Modulation of Iron-Sulfur Interplay in *Arabidopsis thaliana*. *International Journal of Molecular Sciences*, 22(18):10166, 2021. doi: 10.3390/ijms221810166.
- [14] Fiona Cunningham, James E Allen, Jamie Allen, Jorge Alvarez-Jarreta, M Ridwan Amode, Irina M Armean, Olanrewaju Austine-Orimoloye, Andrey G Azov, If Barnes, Ruth Bennett, Andrew Berry, Jyothish Bhai, Alexandra Bignell, Konstantinos Billis, Sanjay Boddu, Lucy Brooks, Mehrnaz Charkhchi, Carla Cummins, Luca Da Rin Fioretto, Claire Davidson, Kamalkumar Dodiya, Sarah Donaldson, Bilal El Houdaigui, Tamara El Naboulsi, Reham Fatima, Carlos Garcia Giron, Thiago Genez, Jose Gonzalez Martinez, Cristina Guijarro-Clarke, Arthur Gymer, Matthew Hardy, Zoe Hollis, Thibaut Hourlier, Toby Hunt, Thomas Juettemann, Vinay Kaikala, Mike Kay, Ilias Lavidas, Tuan Le, Diana Lemos, José Carlos Marugán, Shamika Mohanan, Aleena Mushtaq, Marc Naven, Denye N Ogeh, Anne Parker, Andrew Parton, Malcolm Perry, Ivana Piližota, Irina Prosovetskaia, Manoj Pandian Sakthivel, Ahamed Imran Abdul Salam, Bianca M Schmitt, Helen Schuilenburg, Dan Sheppard, José G Pérez-Silva, William Stark, Emily Steed, Kyösti Sutinen, Ranjit Sukumaran, Dulika Sumathipala, Marie-Marthe Suner, Michal Szpak, Anja Thormann, Francesca Floriana Tricomi, David Urbina-Gómez, Andres Veidenberg, Thomas A Walsh, Brandon Walts, Natalie Willhoft, Andrea Winterbottom, Elizabeth Wass, Marc Chakiachvili, Bethany Flint, Adam Frankish, Stefano Giorgetti, Leanne Haggerty, Sarah E Hunt, Garth R Iisley, Jane E Loveland, Fergal J Martin, Benjamin Moore, Jonathan M Mudge, Matthieu Muffato, Emily Perry, Magali Ruffier, John Tate, David Thibert, Stephen J Trevanion, Sarah Dyer, Peter W Harrison, Kevin L Howe, Andrew D Yates, Daniel R

- Zerbino, and Paul Flicek. Ensembl 2022. *Nucleic Acids Research*, 50(D1):D988–D995, 2021. doi: 10.1093/nar/gkab1049.
- [15] Dmitry Kuznetsov, Fredrik Tegenfeldt, Mosè Manni, Mathieu Seppey, Matthew Berkeley, Evgenia V Kriventseva, and Evgeny M Zdobnov. OrthoDB v11: annotation of orthologs in the widest sampling of organismal diversity. *Nucleic Acids Research*, 51(D1):D445–D451, 2022. doi: 10.1093/nar/gkac998.
- [16] Steven Xijin Ge, Dongmin Jung, and Runan Yao. ShinyGO: a graphical gene-set enrichment tool for animals and plants. *Bioinformatics*, 36(8):2628–2629, 2019. doi: 10.1093/bioinformatics/btz931.
- [17] Daniel C Berrios, Jonathan Galazka, Kirill Grigorev, Samrawit Gebre, and Sylvain V Costes. NASA GeneLab: interfaces for the exploration of space omics data. *Nucleic Acids Research*, 49(D1):D1515–D1522, 2020. doi: 10.1093/nar/gkaa887.
- [18] Ambra S. Parmagnani, Giuseppe Mannino, and Massimo E. Maffei. Transcriptomics and Metabolomics of Reactive Oxygen Species Modulation in Near-Null Magnetic Field-Induced *Arabidopsis thaliana*. *Biomolecules*, 12(12):1824, 2022. doi: 10.3390/biom12121824.
- [19] Giuseppe Mannino, Francesco Caldo, and Massimo E. Maffei. Near-null magnetic fields remodel flavonoid profiles and essential oil content in *Ocimum basilicum* and are associated with altered photochemical performance and ROS homeostasis. *Journal of Plant Physiology*, 326:154872, 2026. doi: 10.1016/j.jplph.2026.154872.
- [20] D. Marshall Porterfield and Richard Barker. Plant Responses to Near-Null Magnetic Fields: Geomagnetism, Bioenergetics, and Primary Metabolism in *Arabidopsis* and *Brassica*. Manuscript in preparation. No DOI has been issued; none is recorded here, deliberately, 2026.